PROGRAMME	:	**Bachelor In Artificial Intelligence with Honours**
COURSE CODE	:	**SAIA2143**
COURSE NAME	:	**Data Wrangling and Visualization**
COURSE LEARNING OUTCOME 2	:	**Construct Data Transformation Using Data Wrangling And Visualization Techniques To Clean And Preprocess Raw Datasets, Addressing Missing Values, Outliers, And Inconsistencies For Analytics**
TAXONOMIES GRADUATE ATTRIBUTES	:	**P4 - Psychomotor Domain Mechanism (Basic Proficiency)**
ASSIGNMENT 1	:	**Group Work (30%)**
MARKS	:	**Group Report in Ms Word (20%)** **Group Presentation in Power Point (5%)** **Individual Role Contribution (5%)**

Case Study: Data Wrangling & Visualization in Python

[40 Marks]

Title: Student Performance Analytics

Dataset Name: Students_Performance_data_set.xlsx

Scenario:
You are working as a **Junior Data Analyst** in the Student Affairs Department at a Malaysian university. The management wants to understand how students' study habits, digital behavior, and lifestyle choices affect their **academic performance (CGPA)**. The department has provided a dataset containing 1,194 student records including information such as:

a. Daily study hours
b. Social media usage
c. English proficiency
d. Co-curricular involvement
e. Attendance

f. Previous SGPA and current CGPA
g. Family income
h. Living situation, and more

You are tasked to perform data wrangling and generate visual insights to support the department in making **data-driven decisions** for academic support programs. Therefore, please answer the question below:

1. Show data information and pre-processing information (profiling and cleaning).

[2 marks]

2. Summarize the group statistics by learning mode in the column: "What is your preferable learning mode?"

   a. Frequency of offline and online
   b. Average CGPA
   c. Count of students per group

[3 marks]

3. Develop a **violin plot** and **box plot** using Seaborn:

   a. X-axis: Learning Mode
   b. Y-axis: Current CGPA
   c. Add color (hue) based on "Gender"
   d. White grid as a theme

[10 marks]

4. Compare on which learning mode has the highest median CGPA?

[5 marks]

5. Compare, is there more variability in CGPA among students who prefer online learning?

[5 marks]

6. Determine the possible factors that could explain the difference in CGPA by learning mode?

[5 marks]

7. Assess TWO (2) the interactivity of a Plotly scatter plot to show how **CGPA** changes across different **learning modes over age or semester level.**

a) **Title = Animated Scatter Plot: Study Hours vs CGPA by Age**
b) **X-axis**: Hours studied per day **and** How many hours do your study daily?

c) **Y-axis**: Another numeric score (e.g., satisfaction level)
d) **Animation frame**: Age group
e) **Size**: Bubble size = "Number of subjects"
f) **Color**: Gender
g) **Height** = 600
h) **Size Max** = 45

[10 marks]

Report Structure Recommendation

A. Formatting Standards

- **Body Text:** Use a clean, professional font.
  - *Recommended:* **Arial**
- **Code Blocks:** Courier New
- **Line Spacing:** 1.5 spacing
- **Alignment:**
  - **Headings:** Left Aligned
  - **Body Text:** Justified

B. Font Sizes

- **Main Title:** 16pt (Bold)
- **Section Headings:** 14pt (Bold)
- **Body Text:** 11pt (Regular)
- **Figure Captions:** 10pt (*Italic*)

Team Roles:

Role	Persona Name	Responsibilities	Key Task
1	Project Lead & Domain Expert	Manages the overall timeline and ensures the analysis aligns with the "Scenario" (Student Affairs Department goals). They are responsible for the final report/presentation narrative.	Interprets the results. Once the Analyst and Visualizer provide the charts and numbers, this person writes the "Data-Driven Decisions" and recommendations for the university management.
2	Data Engineer	Responsible for the initial setup. They handle loading the Students_Performance_data_set.xlsx file, checking for missing values, cleaning column names, and handling data types.	Focuses on **Question 1** (Show data information). They ensure the dataframe is clean and ready for the analyst and visualizer to use without errors.
3	Statistical Analyst	Focuses on numerical aggregations and grouping. They are the "math" person of the group.	Focuses on **Question 2**. They write the Pandas code to groupby the "preferable learning mode" and calculate the required metrics (Frequency, Average CGPA, Count).
4	Visualization Specialist	The "artist" of the group. They are responsible for selecting the right color palettes and ensuring the charts are readable and meet the	Focuses on **Question 3**. They write the Seaborn code for the Violin and Box plots, ensuring the hue="Gender"

		specific requirements (themes, axes labels).	and "White grid" theme are applied correctly.
5	Quality Assurance & Documentation (Optional 5th Member)	If the group has 5 members, this person compiles the Jupyter Notebook or Python script. They ensure code comments are clear, the formatting is consistent, and they double-check that the visualizer's charts match the statistics generated by the analyst.	

PowerPoint

Slides should following the information:

- Slide 1: Group name with roles
- Slide 2: Overview of messy data issues found.
- Slide 3: Techniques applied for data profiling.
- Slide 4: Techniques applied for data cleaning; please mention what are the functions used.
- Slide 5-14: Answer all the questions.
- Slide 15: Lessons learned and reflections for every team roles (1 page).

Presentation

Date: Week 15 [12-16 Jan 2026], class session.
Time: 20-25 minutes per group with Q/A.

Grading Rubric for Group Presentation (5%)

Criteria	5 points (Excellent)	3–4 points (Good)	1–2 points (Needs Improvement)
Clarity & Organization	Clear, logical flow; well-structured slides; smooth delivery	Mostly clear, some minor disorganization	Unclear flow; disorganized or confusing presentation
Use of Visuals & Examples	Effective visuals (snapshots) that support key points	Some visuals used, but not fully integrated	Few or no visuals; little connection to key points
Team Engagement & Participation	All members actively participate and contribute in presentation	Most members participate; some uneven contributions	Only a few members present; limited team effort

Grading Rubric for Individual Role Contribution (5%)

Role	5 points (Excellent)	3–4 points (Good)	1–2 points (Needs Improvement)
1. Project Lead	Efficiently managed the timeline. The final Conclusion is insightful, connects back to the objective, and offers specific, actionable recommendations for the university.	Managed the team well. The conclusion explains the charts but lacks deep insight or specific recommendations.	Poor coordination; team members were confused. The conclusion is missing, superficial, or simply repeats numbers without context.

2. Data Engineer	Dataset is spotless. Successfully handled all specific issues (e.g., "Online" vs "online", Nulls in Income). Code for cleaning is efficient and documented.	Data is mostly clean but missed minor issues (e.g., a typo remains). The dataset was usable for the analyst but required minor fixes.	Significant errors remain (e.g., duplicates not removed). This caused errors for the Analyst and Visualizer downstream.
3. Statistical Analyst	Calculations are 100% accurate. The summary table correctly shows Frequency, Average CGPA, and Counts. Used correct Pandas syntax and formatting is clean.	Calculations are generally correct, but the output formatting is messy (e.g., unrounded decimals like 3.5555). Logic is sound.	Wrong aggregation used (e.g., Sum instead of Mean). The code throws errors, or the numbers are clearly impossible (e.g., CGPA > 4.0).
4. Visual Specialist	Perfect execution. Created both Violin and Box plots. Applied hue="Gender" and whitegrid theme correctly. Labels, titles, and legends are clear and readable.	Generated the plots, but missed a requirement (e.g., forgot the white grid or the hue). The chart is accurate but hard to read (labels too small).	Failed to create the specific plots (e.g., used a Bar chart instead of Violin). The visualization is cluttered, misleading, or crashes the code.
5. QA & Documentation	Code runs perfectly from start to finish. Comments explain why code was written. The Report/PPT formatting is consistent (fonts, spacing) and professional.	The code runs, but comments are sparse. The formatting of the report/PPT is decent but has minor inconsistencies (e.g., mixed fonts).	Code is broken or disorganized. The final submission looks like 5 separate files pasted together with different styles.

The submission to e-learning: 11 January 2026 (11.59pm). Report (.docx, .pdf) and Slide (.pptx) need to be uploaded to e-learning.

Group Member

NO	NAME	GROUP	ROLE
1	ALIA WAJEEHA BINTI KHAIRUL HIZAM	1	5
2	AREESHA NABILA BINTI DICK HILMI	1	4
3	CHENG HAO WEN	1	3
4	FADZLEE ADAM BIN MOHD NAZLEE	1	2
5	FAQIHAH HUMAIRA' BINTI MUHAMMAD FIRHAT	1	1
6	**HO YU XIANG**	**2**	**1**
7	**JAGADESH**	**2**	**5,2**
8	**KIRTHIGGAN A/L SARAVANAN**	**2**	**4**
9	**LAILA SUHAILA BINTI MOHD FARIZOL**	**2**	**3**
10	**MUHAMMAD IMAN HAKIMI BIN ABU SUPIAN**	**2**	**2**
11	MUHAMMAD LUKMAN BIN NASRUM	3	5
12	MUHAMMAD SAFRUL BIN ROSLI	3	1
13	MUHAMMAD ZAHIN BIN MOHD ZAMRI	3	2
14	MUN WENG YANN	3	3
15	NUR NADSYUHA BINTI MUSTAFA	3	4
16	**OSAMA ALCHIKH WARAK**	**4**	**3**
17	**RAQIB HAZIM BIN ABDUL HAMID**	**4**	**1,5**
18	**RYAN ANG JUN YAU**	**4**	**2**
19	**SANG YEN LING**	**4**	**4**
20	SAVEETHA SHREE A/P SURESH	5	4
21	TEOH JAY RONG	5	3
22	VIROCCHAN A/L S.ELANKOVEN	5	1
23	WAN NUR HAJAR BINTI WAN AZLAN	5	2

Notes: If your group has 4 or 5 students